Examen final 2019

1 ΔVS

$A = USV^t$, atunci A^{-1} există $\Leftrightarrow$ S^{-1} există

SVD pt A^{-1} ?

fie $A \in \mathbb{R}^{n \times n}$ cu $A = USV^t$ unde U, V sunt matrice ortogonale și $S = \operatorname{diag}(\sigma_1, \dots \sigma_n)$, $\sigma_i > 0$

($\Rightarrow U$ și V sunt inversabile $\det U$ și $\det V \neq 0$)

PP A inv $\Rightarrow \det A \neq 0$

$\Rightarrow \det U \neq 0$ $\det V \neq 0 \Rightarrow \det V^T \neq 0$ } $\Rightarrow \det S \neq 0$

$\Rightarrow S$ inv

PP S inv $\Rightarrow \det S \neq 0$

$\Rightarrow \det U \neq 0$ $\det V \neq 0 \Rightarrow \det V^T \neq 0$ } $\det A \neq 0$

$\Rightarrow A$ inv

$\Rightarrow$ (∃) $A^{-1} \Leftrightarrow$ (∃) S^{-1}

$A = USV^T$

$A A^{-1} = USV^T A^{-1}$

$V S U^T = A^{-1}$

$\Rightarrow A^{-1} = V S^{-1} U^T$ $\quad S^{-1} = \begin{bmatrix} \frac{1}{\sigma_1} & & \\ & \frac{1}{\sigma_2} & \\ & & \frac{1}{\sigma_n} \end{bmatrix}$

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3. $\int_0^1 f(x)\,dx = c_0(f(0) + f(x_1)) + c_1 f(x_1) = \frac{1}{4}(f(0) + f(\frac{2}{3}) + \frac{1}{2} f(\frac{2}{3})$

grad valabilitate $2n+1$

$f: 1, x, x^2, \ldots$ verifică și pentru $x^3, \ldots$ până nu mai merge

$f(x) = 1$

$\int_0^1 1\,dx = 1 \qquad 1 = c_0(1+1) + c_1$

$1 = 2c_0 + c_1$

$f(x) = x$

$\int_0^1 x\,dx = \frac{1}{2} \qquad \frac{1}{2} = c_0 x_1 + c_1 x_1 = x_1(c_0 + c_1)$ (1)

$f(x) = x^2$ $\qquad x_1 \neq 0$

$\int_0^1 x^2\,dx = \frac{1}{3} \qquad \frac{1}{3} = c_0 x_1^2 + c_1 x_1^2 = x_1^2(c_0 + c_1)$ (2)

$$\left.\begin{aligned} c_0 + c_1 &= \frac{1}{2x_1} \\ c_0 + c_1 &= \frac{1}{3x_1^2} \end{aligned}\right\} \; x_1 = \frac{2}{3}$$

$$\left.\begin{aligned} c_0 + c_1 &= \frac{3}{4} \\ 2c_0 + c_1 &= 1 \end{aligned}\right\} \rightarrow \quad c_0 = \frac{1}{4} \qquad c_1 = \frac{1}{2}.$$

$f(x) = x^3$

$\int_0^1 x^3\,dx = \frac{1}{4} \qquad \frac{1}{4} \neq \frac{3}{4}\left(\frac{2}{3}\right)^3$

$\Rightarrow$ grad 2

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Pade $N=2$

$$f(x) = \sum_{i} (-x)^i / i! = e^{-x}$$

$$f(x) \simeq r(x) = \frac{p(x)}{q(x)} = \frac{p_0 + p_1 x + \dots + p_u x^u}{q_0 + q_1 x + \dots + q_w x^w}$$

$N = u + w = 2$

$$\begin{cases} u=0 \\ w=2 \end{cases} \quad \begin{cases} u=1 \\ w=1 \end{cases} \quad \begin{cases} u=2 \\ w=0 \end{cases}$$

$$f(x) - r(x) = f(x) - \frac{p(x)}{q(x)} = \frac{f(x)\,q(x) - p(x)}{q(x)}$$

(1) $u=0$, $w=2$ — prin simplificare cu q_0

$$r(x) = \frac{p_0}{1 + q_1 x + q_2 x^2}$$

$f(x)q(x) - p(x)$ să aibă $1, x, x^2$ cuf 0

$$\sum_{i} (-x)^i / i! \cdot (1 + q_1 x + q_2 x^2) - p_0 = 0 + 0x + 0x^2$$

$$\left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots\right)(1 + q_1 x + q_2 x^2) - p_0 = 0 + 0x + 0x^2$$

$1 - p_0 = 0 \quad p_0 = 1$

$q_1 - 1 = 0 \quad q_1 = 1$

$q_2 - q_1 + \frac{1}{2} = 0 \quad q_2 = \frac{1}{2}$

$$r(x) = \frac{1}{1 + x_1 + \frac{x_1}{2}}$$

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5 $f(x) = \log_{10} x = \lg x$ pe $[1, 10]$

$\varepsilon = 10^{-6}$

$S = \{(x_0, y_0) \dots (x_n, y_n)\}$

$$|f(x) - P_n(x)| = e(x) = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i) \right|$$

$n = 2$

$$|f(x) - P_2(x)| = \left| \frac{f^{(3)}(\xi)}{6} (x - x_0)(x - x_1)(x - (x_0 + h)) \right|$$

$x_1 \in (x_0, x_0 + h)$

$$= \left| \frac{1}{3 \ln 10} \cdot \frac{1}{\xi^3} (x - x_0)(x - x_1)(x - x_0 - h) \right|$$

$$\leq \frac{1}{3 \ln 10} \left| (x - x_0)(x - x_1)(x - x_0 - h) \right|$$

$$\begin{cases} f(x) = \lg x = \dfrac{\ln x}{\ln 10} \\ f^{(3)}(x) = \dfrac{2}{x^3 \ln 10} \end{cases}$$

$u = x - x_0$ $\quad u, v \in [0, h]$

$v = x_1 - x_0$

$$= \frac{1}{3 \ln 10} \left| u(u - v)(u - h) \right| = \frac{1}{3 \ln 10} u(h - u) |u - v|$$

u maximizează pt $\frac{h}{2}$

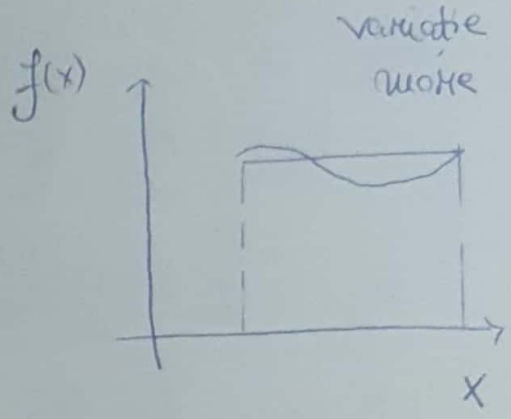
$|u - v|$ este mult h

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$\leq \frac{1}{3 \ln 10} \frac{h^2}{4} h < \varepsilon$

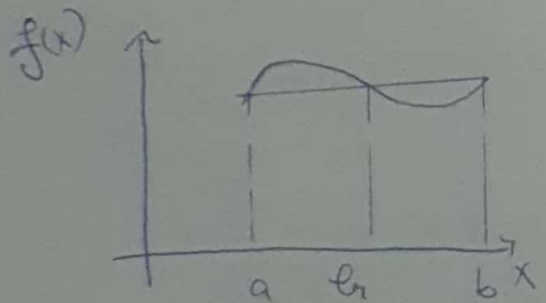
$h^3 < 12 \ln 10 \, \varepsilon$

$h < \sqrt[3]{12 \ln 10 \, \varepsilon}$

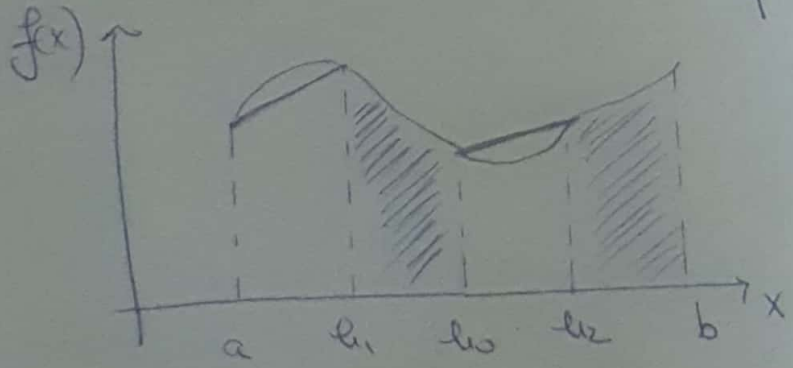


6 explicați grafic cuadraturile adaptive

urmărind să reținem un de pași ocolo unde f variază mult

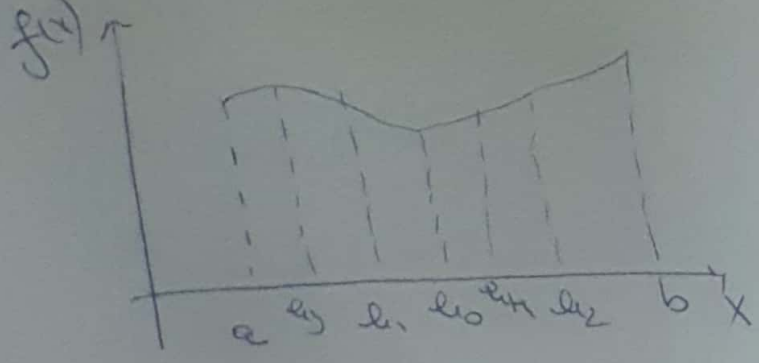


variație mare în anumite intervale



$\notin [h_1, h_0] \cup [h_2, b]$

variația este acceptabilă



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```
function [T] = trapezoidal (f, a, b, n)
    h = (b-a)/n;
    x = (a + h).h.(b - h)
    T = h * (( f(a) + f(b) ) /2 + sum( f(x) ));
endfunction

function [S] = Romberg( f, a, b, n)
    S = zero (n+1, 1);

    for i = 1 : n+1;
        S(i) = trapezoidal( f, a, b, 2^(i-1));
    endfor
    for j = 2 : n+1
        len = length(S);
        prev_S = S;
        S = (4^(j-1) * prev_S(2: len) -
    - prev_S(1 : len-1)) / (4^(j-1) - 1);
    endfor
endfunction
```

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9,414

```
function[a] = cubic_approx(x,y)
    S = zeros(1,9);
    for i = 1:9
        S(i) = sum(x.^(i-1));
    endfor
    b = zeros(4,1);
    for i = 1:4
        b(i) = sum(y.*x.^(i-1));
    endfor
    A = zeros(4);
    for i = 1:4
        for j = 1:4
            A(i,j) = S(i+j-1);
        endfor
    endfor
    a = A\b;
endfunction
```

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```
function Σ = Euler(f, a, b, y0, N)

    h = (b-a)/N
    y = zeros(1, N+1);

    T = a:h:b

    y(1) = y0 ;

    for j = 1:N
        y(j+1) = y(j) + h * f(T(j));

    endfor

    Σ = [T' y']

endfunction

function [Mes] = Lagrange(x, y, q)
    U = eye(n) + ones(n) diag(x) - diag(x) ones(n)
    V = eye(n) + diag(a-x) (n eye(n))
    L = prod(U) / prod(V);
    Mes = L'*y
endfunction
```

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```
function [a] = Quadric_approx (x,y)

    S = zeros(1,5);

    for i = 1:5
        S(i) = sum(x .^ (i-1));
    end for

    b = zeros(3,1);

    for i = 1:3
        b(i) = sum(y .* (x .^ (i-1)));
    end for

    A = zeros(3);

    for i = 1:3
        for j = 1:3
            A(i,j) = S(i+j-1);
        end for
    end for

    a = A \ b;

end function
```

5,3,3

15

3,1

3

(x^{i-1})

$S(i+j-1)$

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